

Question # 1 of 35 (Start time: 06:04:35 AM, 04 July 2020)

The inverse exist only for non-singular matrices.

Select the correct option

☒ True

☐ False

Question # 2 of 35 (Start time: 06:05:47 AM, 04 July 2020)

Given a square matrix A, if there exists a matrix B such that $AB = BA = I$, then A is called the _____ of B

Select the correct option

☒ Multiplicative inverse

☐ Adjoint

☐ Transpose

☐ Additive inverse

Grand Quiz

Question # 3 of 35 (Start time: 06:07:00 AM, 04 July 2020)

If $|A| = 3$ and $\text{adj}(A) = \begin{bmatrix} 9 & -3 \\ -5 & 2 \end{bmatrix}$, then $A^{-1} =$

Choose the correct option

☐ does not exist

☒ $\begin{bmatrix} 3 & -1 \\ \frac{1}{3} & \frac{1}{3} \end{bmatrix}$

☐ $\begin{bmatrix} 3 & -1 \\ 0 & \frac{1}{3} \end{bmatrix}$

☐ none

MTH100: Grand Quiz

Question # 4 of 35 (Start time: 06:08:01 AM, 04 July 2020)

If $f(x)=x$ and $g(x)=1-x$, then the product $(f \cdot g)(x) =$ _____

Select the correct option

☒ $x \cdot x \cdot x$

☐ $x \cdot x \cdot x$

☐ $x \cdot x$

☐ $1 \cdot x \cdot x$

If $A = \{1, 3, 5, 7, 9\}$ $B = \{2, 4, 6, 8, 10\}$ $C = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ then what is $A \cap B \cap C = ?$

Select the correct option

- ☐ $\{1, 3, 5, 7, 9\}$
- ☐ $\{2, 4, 6, 8, 10\}$
- ☐ $\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$
- ☒ $\emptyset$

MTH100: Grand Quiz

Question # 6 of 35 (Start time: 06:09:53 AM, 04 July 2020)

If A and B are matrices, then which of the following is true?

Select the correct option



$$A + B \neq B + A$$



$$AB \neq BA$$

Question # 7 of 35 (Start time: 06:10:28 AM, 04 July 2020)

0 8

is

Select the correct option

0



-8



64



1



MTH100: Grand Quiz

Question # 9 of 35 (Start time: 06:12:34 AM, 04 July 2020)

Which of the following is the common ratio r of a geometric sequence 1, 3, 9, 27, 81, ...?

Select the correct option

☒ 3

☐ $\frac{1}{3}$

☐ -3

☐ 4

MTH100: Grand Quiz

Question # 10 of 35 (Start time: 06:13:17 AM, 04 July 2020)

Which of the following is the Complex conjugate of complex number: $z = (1/i)$?

Select the correct option



$(1/i)$



$(-1/i)$



i



$-i$

Question # 11 of 35 (Start time: 06:14:17 AM, 04 July 2020)

The number $0.3333\dots$ is

Select the correct option

☒ rational

☐ irrational

Question # 12 of 35 (Start time: 06:14:44 AM, 04 July 2020)

Find the FALSE statement.

Select the correct option



Numbers in a sequence always get bigger



A series is a summed list of numbers following some pattern



A sequence is a list of numbers following some pattern



Sequences can be finite or infinite

Question # 13 of 35 (Start time: 06:15:51 AM, 04 July 2020)

For a geometric sequence, if 2nd term is equal to 4th term then value of common ratio is —

Select the correct option

-1 only



1 only



0



+1 or -1



Question # 15 of 35 (Start time: 06:18:50 AM, 04 July 2020)

A Square matrix with ones on the diagonal and zeros elsewhere is called a/an matrix

Select the correct option

☐ Null

☐ scalar

☐ diagonal

☒ identity

QUESTION # 17 of 35 (Start time: 06:19:34 AM, 04 July 2020)

Which of the following number System will have the solution of equation: $x^2 + 6 = 0$

Select the correct option

☐	Integer
☐	Irrational
☒	Complex
☐	Real

MTH100:Grand Quiz

Question # 18 of 35 (Start time: 06:20:10 AM, 04 July 2020)

For an arithmetic sequence with first term = 1, if the sum of its '101' terms is also '101' then the common difference of this AP is —.

Select the correct option

101



100



0



1



MTH100:Grand Quiz

Question # 19 of 35 (Start time: 06:21:40 AM, 04 July 2020)

Cardinalities of $\{1,2,3\}$ and $\{2,1,3\}$ are_____

Select the correct option

Equal



Not equal



Question # 20 of 35 (Start time: 06:22:01 AM, 04 July 2020)

If $A = \{1, 2, 3\}$ and $B = \{2, 3, 4\}$ then ____

Select the correct option



A is subset of B



B is subset of A



Union of A and B has 4 elements



Intersection of A and B has 3 elements

Question # 21 of 35 (Start time: 06:22:40 AM, 04 July 2020)

Which of the following is the correct formula for finding the 4th term in a geometric sequence?

Select the correct option



$$a_4 = ar^5$$



$$a_4 = ar^3$$



$$a_4 = ar$$



none of the above

Question # 22 of 35 (Start time: 06:23:17 AM, 04 July 2020)

The Graph of function: $f(x) = 1 - x \cdot x$ would be a——.

Select the correct option



Circle



Parabola



Hyperbola



Straight line

MTH100: Grand Quiz

Question # 23 of 35 (Start time: 06:23:55 AM, 04 July 2020)

If the first term $= a = 1$, common ratio $= r = 1/2$, then which term of geometric sequence is $1/4$?

▶ Select the correct option

☐

5

☐

6

☒

3

☐

4

MTH100: Grand Quiz

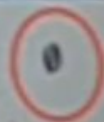
Question # 25 of 35 (Start time: 06:26:28 AM, 04 July 2020)

$$A \cap \emptyset = ?$$

Select the correct option



A



Universal Set



None of these

MTH100: Grand Quiz

Question # 26 of 35 (Start time: 06:27:00 AM, 04 July 2020)

Which of the following matrix has an inverse?

Select the correct option

☒ Singular matrix

☐ Non-singular matrix

MTH100:Grand Quiz

Question # 27 of 35 (Start time: 06:27:24 AM, 04 July 2020)

Generally the matrices are denoted by

Select the correct option



capital letters



numbers



small letters



operational signs

Question # 28 of 35 (Start time: 06:27:43 AM, 04 July 2020)

$$2^1 + 2^2 + 2^3 + \dots + 2^n =$$

Select the correct option

- | | |
|----------------------------------|-------------------|
| ☒ | $2(2^n - 1)$ |
| ☐ | $2(2^{n-1} - 1)$ |
| ☐ | $2(2^{n+1} - 1)$ |
| ☐ | none of the above |

Question # 29 of 35 (Start time: 06:29:07 AM, 04 July 2020)

For an Arithmetic series, if $a_1 = -3$, $d = 2$ and $S_n = 21$. Then the number of terms " n " in the series are——.

Select the correct option

☒

7

☐

-7

☐

10

☐

5

Question # 30 of 35 (Start time: 06:30:38 AM, 04 July 2020)

$\sqrt{22/7}$ is an example of _____ number.

Select the correct option

☐

Integer

☐

Natural

☐

Rational

☒

Irrational

Question # 31 of 35 (Start time: 06:30:50 AM, 04 July 2020)

In a system of linear equations, is the determinant of the coefficients of the variables is zero?

Select the correct option

yes



no



MTH100: Grand Quiz

Question # 32 of 35 (Start time: 06:31:41 AM, 01 July 2020)

$f(x) = x + 2$ is an example of _____

Select the correct option

☐	1-1 function
☐	Invertible function(having an inverse)
☐	Linear Function
☒	All above choices equally holds

Question # 33 of 35 (Start at: 06:32:35 AM, 04 July 2020)

If B is a subset of A and A contains elements which are not in B then

Select the correct option



B is improper subset of A



A is improper subset of B



A is proper subset of B



B is proper subset of A

Question # 34 of 35 (Start time: 06:33:11 AM, 04 July 2020)

The explicit formula when $a_1 = 6$ and $d = 2$ is _____

Select the correct option



$$a_n = 6 + 2(n + 1)$$



$$a_n = 6 + 2n$$



$$a_n = 6 - 2(n - 1)$$



$$a_n = 6 + 2(n - 1)$$

Question # 35 of 35 (Start time: 06:34:18 AM, 04 July 2020)

A pair of equations to determine the value of 2 variables is called

Select the correct option

☒ simultaneous linear equations

☐ paired equations

☐ quadratic equations

☐ simple equations

MTH100 Grand Quiz

Question # 1 of 35 (Start time: 07:52:33 AM, 04 July 2020)

6C_2 is equal to

Select the correct option



14



15



16



17

Question # 2 of 35 (Start time: 07:53:18 AM, 04 July 2020)

If $C_n = n(n+1)$ for all integers $n \geq 1$ then which of the following is the first three terms of a sequence?

Select the correct option



0, 2, 6...



2, 6, 12...



1, 2, 3...



1, 2, 6...

Question # 3 of 35 (Start time: 07:54:45 AM, 04 July 2020)

For the Complex numbers: " $z = i$ and $w = (1/i)$ ", the sum: $z+w =$ -----.

Select the correct option



$1+0i$



$2i$



$-2i$



$0+0i$

Question # 4 of 35 (Start time: 07:55:42 AM, 04 July 2020)

Which of the following is the Imaginary part of complex number: $z = i$?

Select the correct option

☐	0
☒	1
☐	-1
☐	None of these

Question # 5 of 35 (Start time: 07:56:13 AM, 04 July 2020)

Two matrices A and B are added if

Select the correct option



both are rectangular



both have same order



no. of columns of A is equal to columns of B



no. of rows of A is equal to no. of columns of B

Question # 6 of 35 (Start time: 07:56:42 AM, 04 July 202

The 9999th term of the sequence:
 $\{-1, 1, -1, 1, -1, 1, \dots\}$ is

▶ Select the correct option

- | | |
|----------------------------------|-------|
| ☒ | -1 |
| ☐ | 9999 |
| ☐ | -9999 |
| ☐ | 1 |

Question # 7 of 35 (Start time: 07:58:11 AM, 04 July 2020)

"1/0" is an example of -----number.

Select the correct option

☐	Rational
☐	Irrational
☐	Complex
☒	None of these

Question # 8 of 35 (Start time: 07:58:47 AM, 04 July 2020)

Two matrices A and B are multiplied to get AB if

Select the correct option



both are rectangular



both have same order



no. of columns of A is equal to no. of rows of B



no. of rows of A is equal to no. of rows of B

Question # 9 of 35 (Start time: 07:59:21 AM, 04 July 2020)

If 1, 2, 3, ... represent a sequence of numbers, then the corresponding series is

Select the correct option



$$1+2+3+\dots$$



$$1.2+2.3+3.4+\dots$$



$$2+3+4+\dots$$



none of the above

Question # 11 of 35 (Start time: 08:01:09 AM, 04 July 2020)

The sum of an infinite geometric series exist only if the condition on the common ration r is

Select the correct option

- | | |
|----------------------------------|--------------------|
| ☐ | $-1 < r < 1$ |
| ☒ | $-1 \leq r \leq 1$ |
| ☐ | $r > -1, r < 1$ |
| ☐ | none of the above |

MTH100: Grand Quiz

Question # 12 of 35 (Start time: 08:02:39 AM, 04 July 2020)

$$A \cap \emptyset = ?$$

Select the correct option

- | | |
|----------------------------------|---------------|
| ☐ | A |
| ☒ | $\emptyset$ |
| ☐ | Universal Set |
| ☐ | None of these |

MTH100: Grand Quiz

Question # 13 of 35 (Start time: 08:03:12 AM, 04 July 2020)

Find the FALSE statement.

Select the correct option



Numbers in a sequence always get bigger



A series is a summed list of numbers following some pattern



A sequence is a list of numbers following some pattern



Sequences can be finite or infinite

Question # 19 of 35 (Start time: 08:08:14 AM, 04 July 2020)

$f(x)=x.x$ is an example of ———.

Select the correct option



1-1 function



Many one function



One many function

Bijective function

Question # 29 of 35 (Start time: 08:17:07 AM, 04 July 2020)

Which of the following is the common ratio r of a geometric sequence 1, 3, 9, 27, 81, ...?

Select the correct option



3



$1/3$



-3



4

Question # 31 of 35 (Start time: 08:17:48 AM, 04 July 2020)

Identify the sequence as Arithmetic, Geometric or neither.
8, 13, 18, 23, ...

Select the correct option

☒	Arithmetic
☐	Geometric
☐	Harmonic
☐	Neither

Question # 32 of 35 (Start time: 08:18:20 AM, 04 July 2020)

If first term of the geometric sequence is 243 and sixth term is 1, then the common ratio will be—.

Select the correct option

☐ 3

☐ 1

☒ $\frac{1}{3}$

☐ -1

MTH100:Grand Quiz

Question # 33 of 35 (Start time: 08:19:33 AM, 04 July 2020)

If the "common ratio = 3" for the following geometric sequence:
 $(x-1), (x-1)(x-1), (x-1)(x-1)(x-1), \dots$, then the value of 'x' is _____

Select the correct option

3



4



2



1



MTH100:Grand Quiz

Question # 34 of 35 (Start time: 08:20:45 AM, 04 July 2020)

The number $0.3333\dots$ is

Select the correct option

☒ rational

☐ irrational

MTH100:Grand Quiz

Question # 35 of 35 (Start time: 08:20:54 AM, 04 July 2020)

The inverse exist only for non-singular matrices.

Select the correct option

True



False





Which of the following is the correct formula for finding the 4th term in a geometric sequence?

Select the correct option



$$a_4 = ar^5$$



$$a_4 = ar^3$$



$$a_4 = ar$$



none of the above



Question # 2 of 35 (Start time: 08:10:37 PM, 04 July 2020)

Which of the following is the common difference d of a arithmetic sequence 1, 4, 7, 10, ...?

Select the correct option

- ☐ 2
- ☒ 3
- ☐ -3
- ☐ -2

Click to Save



Question # 3 of 35 (Start time: 08:10:45 PM, 04 July 2020)

If $C_n = n(n+1)$ for all integers $n=1$ then which of the following is the first three terms of a sequence?

Select the correct option



0, 2, 6...



2, 6, 12...



1, 2, 3...



1, 2, 6...

Click to Solve

MTH100: Grand Quiz



Question # 4 of 35 (Start time: 08:11:14 PM, 04 July 2020)

If a scalar 2 is multiplied with the matrix then which one of the following statement is correct?

Select the correct option

- ☒ 2 will be multiplied with all elements of the matrix
- ☐ 2 will be multiplied with first column in the matrix
- ☐ 2 will be multiplied with first row in the matrix
- ☐ 2 will be multiplied with the diagonal elements

Click to Save



Question # 5 of 35 (Start time: 08:11:32 PM, 04 July 2020)

For a sequence whose n th term is $\{-200/n\}$ then its 200th term is -----.

Select the correct option

- | | |
|----------------------------------|-------|
| ☐ | 200 |
| ☐ | 1 |
| ☒ | -1 |
| ☐ | -4000 |

Click to Save



Question # 6 of 35 (Start time: 08:11:44 PM, 04 July 2020)

Determinant of identity matrix is

Select the correct option

- | | |
|----------------------------------|---|
| ☒ | 1 |
| ☐ | 0 |
| ☐ | 2 |
| ☐ | 3 |

Click to Save



Question # 7 of 35 (Start time: 08:11:54 PM, 04 July 2020)

Two matrices of same order can be

Select the correct option



added but not subtracted



subtracted but not added



added and subtracted



multiplied

Click to Save



Question # 8 of 35 (Start time: 08:12:07 PM, 04 July 2020)

If the sum of the infinite geometric series is double of the first term of the series, then the common ratio will be-----.

Select the correct option

- ☐ 1
- ☒ 2
- ☐ 1/3
- ☐ 1/2

Click to S

Question # 9 of 35 (Start time: 08:12:22 PM, 04 July 2020)

If $f(x)=x-1$ and $g(x)=1-x$, then the composition $(f \circ g)(x) =$ -----

Select the correct option



1



-1



-x



x



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Question # 10 of 35 (Start time: 08:12:45 PM, 04 July 2020)

If B is a matrix whose determinant is zero then its inverse

Select the correct option



will be zero



cannot be determined



will be negative



will be positive

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Question # 11 of 35 (Start time: 08:13:01 PM, 04 July 2020)

If B is a subset of A and A contains elements which are not in B then

Select the correct option



B is improper subset of A



A is improper subset of B



A is proper subset of B



B is proper subset of A



Question # 12 of 35 (Start time: 08:13:22 PM, 04 July 2020)

$$\begin{vmatrix} 1 & 5 & -7 \end{vmatrix}$$

is

Select the correct option

- ☒ -28
- ☐ 46
- ☐ 0
- ☐ 4

Question # 13 of 35 (Start time: 08:14:54 PM, 04 July 2020)



If first term of the geometric sequence is 243 and sixth term is 1, then the common ratio will be----

Select the correct option

☐	3
☐	1
☒	1/3
☐	-1



Click to Sa



Question # 14 of 35 (Start time: 08:16:14 PM, 04 July 2020)

If the order of matrix A is 2×3 then what should be the order of matrix B such that addition of matrices A and B is possible.

Select the correct option

- ☐ 3×5
- ☐ 3×2
- ☒ 2×3
- ☐ 2×2





Question # 15 of 35 (Start time: 08:16:30 PM, 04 July 2020)

If $f(x)=x$ and $g(x)=1-x$, then $(f \circ g)(x) = \dots\dots\dots$.

Select the correct option

- | | |
|----------------------------------|-----------|
| ☐ | 1 |
| ☒ | $x(1-x)$ |
| ☐ | $(1-x)/x$ |
| ☐ | $1-(1/x)$ |

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Question # 16 of 35 (Start time: 08:16:39 PM, 04 July 2020)

Sum of first 999 terms of the sequence:
 $\{-1, 1, -1, 1, \dots\}$ is —.

Select the correct option

☐	0
☐	1
☒	-1
☐	Undefined

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Question # 17 of 35 (Start time: 08:17:17 PM, 04 July 2020)

Cardinality of the Union of sets $\{ \}$ and $\{ \{ \}$ are.....

Select the correct option

- | | |
|----------------------------------|---------------|
| ☐ | 2 |
| ☒ | 1 |
| ☐ | 0 |
| ☐ | None of these |

Click to Save



Question # 18 of 35 (Start time: 08:17:27 PM, 04 July 2020)

A pair of equations to determine the value of 2 variables is called

Select the correct option

- | | |
|----------------------------------|-------------------------------|
| ☒ | simultaneous linear equations |
| ☐ | paired equations |
| ☐ | quadratic equations |
| ☐ | simple equations |



Question # 19 of 35 (Start time: 08:17:53 PM, 04 July 2020)

Which of the following is the imaginary part of complex number: $z = (1/i)$?

Select the correct option

- | | |
|----------------------------------|---------------|
| ☐ | 0 |
| ☐ | 1 |
| ☒ | -1 |
| ☐ | None of these |

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Question # 20 of 35 (Start time: 08:18:18 PM, 04 July 2020)

What is the 31st term of the arithmetic sequence: 1, 4, 7, 10, ...?

Select the correct option

- | | |
|----------------------------------|----|
| ☐ | 90 |
| ☒ | 91 |
| ☐ | 92 |
| ☐ | 93 |

Click to Save



Question # 21 of 35 (Start time: 08:18:49 PM, 04 July 2020)

$f(x) = 1-x$ is an example of -----.

Select the correct option

- | | |
|----------------------------------|-------------------------------|
| ☐ | Even function |
| ☐ | Odd function |
| ☐ | Both even and Odd function. |
| ☒ | Neither even nor Odd function |



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Question # 22 of 35 (Start time: 08:19:10 PM, 04 July 2020)

Sum of the first 20 terms of the following series:

$1+2+3+4+\dots$ is-----

Select the correct option

55

☐

200

☐

220

☐

210

☒

Click to Save



1 1 3

is

Select the correct option

- ☒ -7
- ☐ 7
- ☐ 8
- ☐ 10



Question # 24 of 35 (Start time: 08:20:33 PM, 04 July 2020)

Which of the following is the Imaginary part of complex number: $z = i$?

Select the correct option

- | | |
|----------------------------------|---------------|
| ☐ | 0 |
| ☒ | 1 |
| ☐ | -1 |
| ☐ | None of these |

Click to Save



Question # 25 of 35 (Start time: 08:20:49 PM, 04 July 2020)

If $X = \{1, 2, 3\}$, and $Y = \{p, q, r\}$ then $X \cup Y$ is equal to

Select the correct option

☐ {1, 2, 3}

☐ {p, q, r}

☐ {1, 2, 3, p}

☒ {1, 2, 3, p, q, r}

Question # 26 of 35 (Start time: 08:20:59 PM, 04 July 2020)

Domain of function: $f(x)=1/(\sqrt{x-1})$ is all Real numbers -----.



Select the correct option

- | | |
|----------------------------------|--------------|
| ☒ | $x > 1$ |
| ☐ | $x < 1$ |
| ☐ | except $x=1$ |
| ☐ | except $x=0$ |



Click to Save



Question # 27 of 35 (Start time: 08:21:58 PM, 04 July 2020)

A Square matrix with ones on the diagonal and zeros elsewhere is called a/an matrix

Select the correct option

- ☐ Null
- ☐ scalar
- ☐ diagonal
- ☒ identity

The graph of function $f(x)=x.x-1$ will have least one ---.



Select the correct option

Minimum



Maximum



Click to Save





Question # 29 of 35 (Start time: 08:22:21 PM, 04 July 2020)

Which of the following expression is true ?

Select the correct option

☐	$N \subset R \subset Z \subset Q$
☐	$R \subset Q \subset Z \subset N$
☒	$N \subset Z \subset Q \subset R$
☐	None of these





Question # 30 of 35 (Start time: 08:22:35 PM, 04 July 2020)

$f(x)=x+2$ is an example of.....

Select the correct option

- ☐ 1-1 function
- ☐ Invertible function(having an inverse)
- ☐ Linear Function
- ☒ All above choices equally holds

Click to Save



Question # 31 of 35 (Start time: 08:22:44 PM, 04 July 2020)

If $f(x)=x-1$ and $g(x)=1-x$, then the Domain of composition $(f \circ g)(x)$ is-----

Select the correct option

- ☐ Set of all Natural numbers
- ☐ Set of Rational numbers
- ☐ Set of all Irrational numbers
- ☒ Set of all Real numbers

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Question # 32 of 35 (Start time: 08:23:04 PM, 04 July 2020)

If $a = 4$, and $d = 5$ in an arithmetic sequence, then what is the value of sixth term?

Select the correct option



25



24



23



29



Click to Save



Question # 33 of 35 (Start time: 08:23:34 PM, 04 July 2020)

Cardinality of the intersection of sets $\{ \}$ and $\{ \{ \}$ is.....

Select the correct option



3



2



1



0

Click to Save



Question # 34 of 35 (Start time: 08:23:45 PM, 04 July 2020)

If the first term= $a=1$, common ratio= $r=1/2$, then which term of geometric sequence is $1/4$?

Select the correct option

- | | |
|----------------------------------|---|
| ☐ | 5 |
| ☐ | 6 |
| ☒ | 3 |
| ☐ | 4 |



MTH100: Grand Quiz

Question # 35 of 35 (Start time: 08:25:05 PM, 04 July 2020)

By changing the signs of all elements of a matrix, we obtained

Select the correct option

☐ identical matrix

☒ negative of a matrix

☐ zero matrix

☐ determinant of a matrix

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